

Component 4 – Production Analysis & Resource Optimization System

Feature Document

1. System Overview

The **Production Analysis & Resource Optimization System** is designed to analyze production performance, compare production plants, optimize resource allocation, and support management decisions.

The system evaluates both existing registered production plants and external/local sub plants. Local sub plants can use a connected **Local Sub Plant Portal** to record their operational data. This information is transferred to the main Component 4 system and used for performance analysis.

The system mainly considers:

- Production efficiency
- Daily production commitment
- On-time order completion
- Damage percentage
- Plant workload
- Resource utilization
- Order quantity
- Production cost/rate
- Historical plant performance

The system can therefore identify high-performing plants and recommend suitable plants for future orders.

2. Main Component 4 Dashboard

The main dashboard provides an overall view of production performance.

Main Dashboard Features

2.1 Average Overall Performance Score

Displays the average performance score of all analyzed plants.

2.2 Best Performing Plant

Automatically identifies the plant with the highest overall performance score.

Example:

Dinusha Embroidery – 4.35 / 5

2.3 Average On-Time Rate

Displays the percentage of production orders completed within the required production dates.

2.4 Average Damage Percentage

Displays the average production damage percentage across plants.

2.5 Total Urgent Orders

Displays the total number of urgent production orders handled by the plants.

3. Plant Performance Ranking

The system ranks plants using production performance indicators.

The ranking table contains:

Feature	Description
Plant Name	Name of the production plant
Overall Score	Overall plant performance score
On-Time Rate	Percentage of orders completed on time
Efficiency	Production efficiency percentage
Damage %	Percentage of damaged items
Rank	Performance position of the plant

Plants with better quality, lower damage, higher efficiency and better on-time production receive higher rankings.

4. KPI Comparison

The system compares plants using important Key Performance Indicators.

Main KPIs include:

- Quality Rating
- On-Time Rate
- Efficiency

- Utilization
- Damage Percentage
- Delay Ratio
- Daily Commitment
- Workload
- Urgent Orders Handled
- Historical Performance

Graphs and charts are used to make comparisons easier for management.

5. Delay and Damage Analysis

This feature identifies plants with production delay and quality problems.

The system analyzes:

- Delay Ratio
- Damage Quantity
- Damage Percentage
- Order Quantity
- Completed Quantity

Damage Percentage Rule

Damage percentage is calculated using:

$$\text{Damage Percentage} = (\text{Damaged Quantity} / \text{Total Order Quantity}) \times 100$$

The normal acceptable damage level is:

$$\text{Damage Percentage} \leq 3\%$$

If the damage percentage is below 3%, the plant performance is considered acceptable.

However:

The lower the damage percentage, the better the plant quality performance.

Example:

Plant	Damage %	Evaluation
Plant A	0.50%	Excellent
Plant B	1.20%	Very Good
Plant C	2.50%	Acceptable
Plant D	3.50%	Needs Improvement

6. Workload and Resource Utilization

This feature evaluates whether production plants are:

- Underutilized
- Optimally utilized
- Overloaded

The system analyzes available production capacity and current workload.

This helps management redistribute orders between plants and improve overall production efficiency.

7. Machine Learning Prediction

Machine learning models are used to improve production analysis and plant recommendations.

The system can compare models such as:

- Random Forest
- Gradient Boosting

Performance measurements may include:

- MAE
- RMSE
- R² Score
- Prediction Accuracy

The best-performing model is selected for prediction and recommendation tasks.

8. High PCS Order – Best Rate Plant Recommendation

This feature is used when a company receives a high quantity production order.

Inputs

- Order Quantity
- Buyer
- Required Date

- Plant Production Rate
- Plant Capacity
- Damage Percentage
- Historical Performance

System Process

The system compares available plants and identifies the plant that offers the best production rate while maintaining acceptable production quality.

Output

The system displays:

- Recommended Plant
- Rate per PCS
- Expected Damage Percentage
- Estimated Completion Time
- Estimated Production Cost
- Recommendation Confidence

The recommendation helps reduce production cost while keeping the expected damage percentage low.

9. Suggested Plant for New Order

Management can enter a new order into the system.

Example inputs:

- Order Quantity
- Required Date
- Priority
- Buyer
- Quality Requirement

The system analyzes plant performance and recommends the most suitable production plant.

Recommendation factors include:

- Available Capacity
- Historical On-Time Performance
- Damage Percentage
- Production Rate
- Daily Commitment
- Current Workload
- Quality Performance

10. Other Plant Optimization

The **Other Plant Optimization** feature evaluates external/local sub plants that are not currently included among the company's major registered production plants.

The objective is to discover new production partners with better performance.

Main Information Displayed

- Plant Name
- Order Started Date
- Location
- Customer Type
- Order Quantity
- Daily Commitment
- Completed Quantity
- Damaged Quantity
- Damage Percentage
- Order Status

Main Recommendations

The system identifies:

Best Daily Commitment

The plant capable of producing the highest quantity per day.

Minimum Damage Percentage

The plant with the lowest production damage percentage.

Best On-Time Completion

Plants that successfully complete production orders within planned dates.

These results allow management to compare local sub plants with existing production plants.

If a local sub plant performs better, management can consider using that plant for future production orders.

11. Local Sub Plant Portal

A separate connected subsystem is provided to local production plants.

Each local sub plant can enter its own operational information.

Example:

Lanka Mount Castle (Pvt) Ltd

The data entered through this portal is connected to the main Component 4 analysis system.

The portal contains the following major modules:

1. Customer Register
2. New Order Register
3. Gatepass
4. Invoice
5. Performance / Analytics

12. Customer Register

This feature allows a local sub plant to register customers.

Input Fields

- Date
- Customer Type
- Customer ID
- Customer Name / Company Name
- Customer Number
- Customer Location

Additional Features

The customer dashboard displays:

- Total Customers
- Active Customers
- New Customers
- Customers with Pending Payments
- Customer Type Distribution
- New Customers Over Time

A customer record table stores previously registered customers.

13. New Order Register

This feature allows the local plant to register new production orders.

Input Fields

- Date
- Order Number
- Customer Name
- Order Quantity
- Planned Start Date
- Planned Finish Date

Dashboard Information

The system can display:

- Orders This Month
- Planned Quantity
- Average Daily Commitment
- On-Time Completion Rate

Graphs

The page can include:

- Order Quantity Trend
- Order Status Distribution

Order Status

Orders can be categorized as:

- Draft
- Confirmed
- In Progress
- Completed

Previous orders are stored in a **Recent Orders** table.

14. Gatepass Management

The Gatepass feature records completed or dispatched production quantities.

Information Recorded

- Gatepass Number
- Date

- Customer
- Style Number
- Description
- Good Quantity
- Damage Quantity
- Total Quantity
- Dispatch Status

This information helps the main system calculate actual production output and damage performance.

15. Invoice Management

The Invoice module manages customer invoices and payments.

Invoice Creation

The system allows the user to enter:

- Date
- Customer
- Invoice Number
- Description
- Amount
- Discount
- Total Price

The user can then:

- Generate Invoice
 - Print Invoice
 - Save Invoice Record
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16. Invoice Financial Dashboard

The Invoice module also provides financial information.

Monthly Income

Displays total income generated during the selected month.

Yearly Income

Displays total income generated during the year.

Paid Amount

Displays the total amount already collected from customers.

Pending Payment

Displays the outstanding amount that has not yet been paid.

17. Payment Status Analysis

The system classifies invoices as:

- Paid
- Partial Paid
- Pending
- Overdue

A graphical representation can display the percentage of paid and pending invoices.

Example:

Paid – 73.3%

Pending – 26.7%

18. Invoice Records

Previously generated invoices are stored in an Invoice Records table.

The table can contain:

- Invoice Number
- Customer
- Invoice Date
- Amount
- Paid Amount
- Balance
- Payment Status
- Date

This allows the local plant to maintain a complete financial transaction history.

19. Local Plant Performance Data Integration

Information entered into the Local Sub Plant Portal is transferred to the main Component 4 system.

Important transferred information includes:

- Order Quantity
- Daily Commitment
- Planned Start Date
- Planned Finish Date
- Actual Completion Date
- Completed Quantity
- Damaged Quantity
- Damage Percentage
- Customer Orders
- Production Rate
- Order Status

The main system analyzes this information and calculates the performance of each local plant.

20. Local Plant Comparison

The system compares local sub plants with existing production plants.

The comparison considers:

- Daily Commitment
- Damage Percentage
- On-Time Completion
- Production Capacity
- Efficiency
- Production Cost
- Historical Performance

If a local plant performs better than an existing production plant, the system can highlight that plant as a potential production partner.

21. Resource Optimization Recommendations

Based on the analysis, the system can provide recommendations such as:

- Allocate more orders to high-performing plants.
 - Reduce orders assigned to overloaded plants.
 - Transfer production to plants with available capacity.
 - Select plants with lower damage percentages.
 - Select plants with better daily commitments.
 - Select plants with better production rates for high PCS orders.
 - Consider high-performing local sub plants for future production.
 - Improve machine utilization in underutilized plants.
 - Provide quality improvement actions for plants with high damage percentages.
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22. Reports and Analytics

The system allows management to view visual reports using:

- Bar Charts
- Line Charts
- Donut Charts
- Performance Cards
- Ranking Tables
- Trend Graphs

Reports can include:

- Plant Performance Report
 - Damage Analysis Report
 - Delay Analysis Report
 - Production Efficiency Report
 - Local Plant Comparison Report
 - Invoice Report
 - Income Report
 - Order Performance Report
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23. Main System Benefits

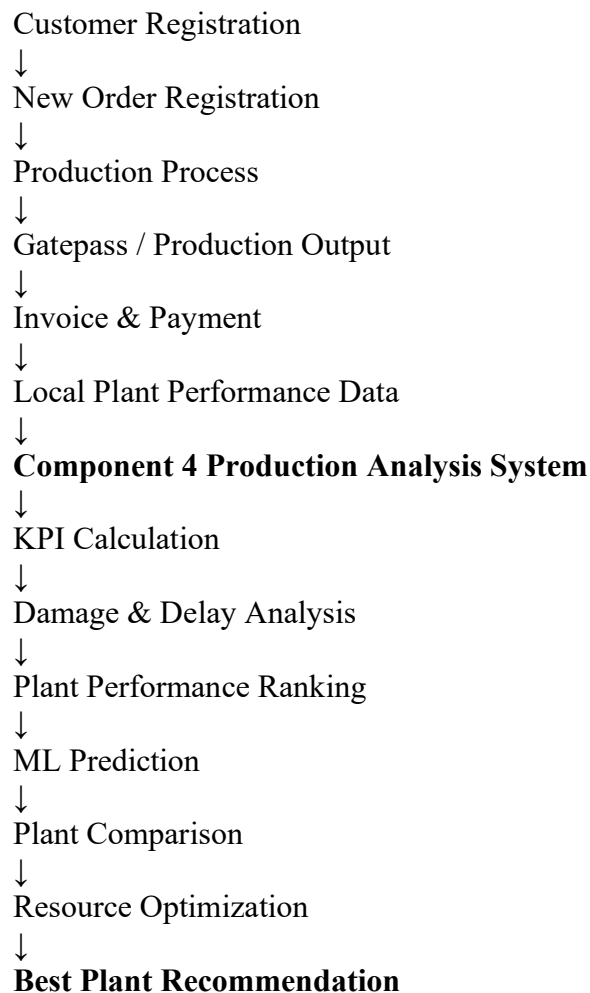
The Component 4 system provides the following benefits:

- Improves production plant selection.

- Reduces production damage.
 - Improves on-time order completion.
 - Identifies high-performing local plants.
 - Reduces production costs.
 - Improves resource utilization.
 - Supports high-volume order allocation.
 - Provides data-driven plant recommendations.
 - Provides better visibility of plant performance.
 - Supports management decision-making.
 - Creates a larger production plant network using external/local sub plants.
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24. Overall System Flow

Local Sub Plant Portal



Conclusion

The Production Analysis & Resource Optimization System provides a centralized method for measuring and improving production plant performance.

The integration of the Local Sub Plant Portal allows the company to collect real production information from external production plants. This enables management to compare existing plants with potential new production partners and identify plants with higher production capacity, lower damage percentages, better daily commitments, lower production rates, and stronger on-time performance.

As a result, the system supports more accurate production allocation and better resource optimization decisions.